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Manual DNA Extraction Protocol for *Campylobacter* Detection in Canine and Bovine Feces

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We use this protocol and it's working

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Disclaimer

Reference to any commercial materials, equipment, or process does not in any way constitute approval, endorsement, or recommendation by the Food and Drug Administration.

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Abstract

Campylobacter jejuni is a leading cause of foodborne illness in people. Animals serve as a reservoir for human infection. While standard methods for the detection of *C. jejuni* in food products are readily available, there is considerable variation in methods published for detection in fecal samples from animals. This method was developed to provide a standard method for the detection of *C. jejuni* in feces from veterinary patients.

Validation data (in-house and by an independent laboratory via collaborative study such as Randomized Blinded Method Test) are available upon request.

Guidelines

All manufacturers' intended use, safety information, storage guidelines, and recommendations should be followed.

Campylobacter jejuni is infectious to humans and is a biological safety level 2 organism. All work with specimens that potentially contain *C. jejuni* should follow the best practices outlined in the most current version of Biosafety in Microbiological and Biomedical Laboratories (BMBL) and institutional guidance.

Reporting of positive test results should follow all local, state, and federal regulations.

Materials

QIAamp® PowerFecal® Pro DNA Isolation Kit (Qiagen, Inc. cat#51804-50)
 VetMAX™ Xeno™ Internal Positive Control DNA (ThermoFisher cat#s A29762/A29764)
 VetMAX™ Xeno™ Internal Positive Control - VIC™ Assay (ThermoFisher Scientific A29765 or A29767)
 TaqMan Fast Virus 1-step Master Mix (ThermoFisher Cat. Nos. 4444432/4444434/4444436)
 Positive control DNA from *C. jejuni* ATCC 33291, ATCC 33292, or ATCC 33560
 Negative control DNA from *E. coli* ATCC 25922
 1X PBS (Life Technologies cat#10010-023)

From any manufacturer:

Microcentrifuge tubes - 1.5 mL, sterile
 50 mL conical tubes - sterile
 Pipette tips - 1 mL
 Pipette tips - 200 µL
 Serological pipette, 25 mL

	A	B
	Oligo Name	5'-3' Sequence
	gyrA-F	AAGATACGGTCGATTTTGTTC
	gyrA-R	CTACAGCTATACCACTTGAACCATTTAATA
	gyrA-P	5'-[FAM]TGATGGTTCAGAAAGCGAACCTGATGTTTT[BHQ1]-3'

Primers and probe for *Campylobacter jejuni* PCR

Primers and probe are diluted to a working concentration of 10 µM with a final concentration in the total reaction volume of 0.5 µM.

Equipment needed:

Scale for weighing 8 g of feces
 Pipettes for appropriate volumes
 Centrifuge for microcentrifuge tubes capable of speeds of up to 16,000 X g
 Vortexer for the fecal slurry
 Vortexer with Qiagen (MoBio) Vortex Adapter (Cat#13000-V1-24)

Reference for primers and probe

Iijima Y, Asako NT, Aihara M, Hayashi K. (2004) Improvement in the detection rate of diarrhoeagenic bacteria in human stool specimens by a rapid real-time PCR assay. J Med Microbiol. 2004 Jul;53(Pt 7):617-22. PMID: 15184531



Protocol materials

⊗ 1X PBS **Life Technologies Catalog #10010-023**

⊗ VetMAX Xeno Internal Positive Control ND **Thermo Fisher Scientific Catalog #A29762/A29764**

⊗ QIAamp PowerFecal Pro DNA Isolation Kit **Qiagen Catalog #51804-50**

Safety warnings

- ! All manufacturers' intended use, safety information, storage guidelines, and recommendations should be followed.

Appropriate safety and personal protective equipment should be worn while working with chemicals including a suitable lab coat, disposable gloves, and protective eyewear.

Campylobacter jejuni is infectious to humans and is a biological safety level 2 organism. Work with specimens that may contain this organism should follow all applicable institutional guidelines.



Background

- 1 Fecal samples should be stored at 4-8 °C refrigerated until processed.
- 2 Samples should be **processed within 72 hours of receipt at the laboratory.**
- 3 Plates and enrichment broth should be incubated in a microaerophilic environment at 42 °C + or - 2 .
- 4 Remaining samples from the associated culture method can be used for molecular detection.
- 5 This protocol uses
 VetMAX Xeno Internal Positive Control ND **Thermo Fisher Scientific Catalog #A29762/A29764**
as a control for PCR inhibition.

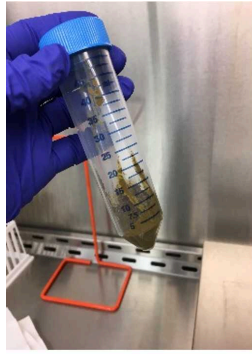
Note

Note: This reagent can either be added at step 15 to serve as a DNA extraction control or after DNA extraction is complete to serve as the PCR inhibition control. If the laboratory has a system in place for DNA extraction and PCR inhibition, that method can be substituted in this protocol.

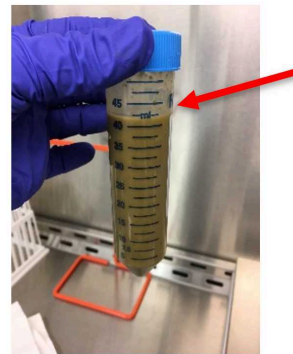
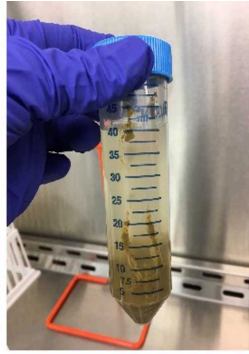
Sample Preparation

5m

- 6 Use 8 g of stool in 50 mL conical tubes
- 7 Add 32 mL 1X PBS **Life Technologies Catalog #10010-023** to 8 g of feces. The feces in PBS will hereafter be called the feces-PBS slurry.



8 g feces



Add **32 mL PBS** to 8 g feces to prepare the feces-PBS slurry and allow the tube to sit for 2 minutes.

Vortex slurry for 30 seconds. Use slurry for direct PCR.

When transferring sample from the feces-PBS slurry pipette from the meniscus at the red arrow.

Making the Feces-PBS slurry.

7.1 Allow samples to sit for 00:02:00 after adding the 32 mL PBS

2m

7.2 Vortex samples for 00:00:30 to prepare fecal slurry

30s

PCR Method - DNA Extraction

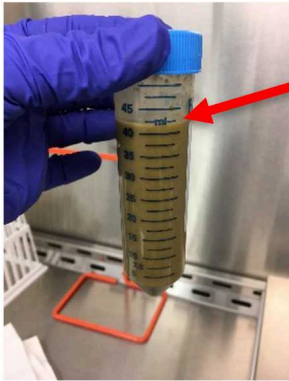
19m 10s

8 Extract DNA from the feces using the QIAamp PowerFecal Pro DNA Isolation Kit **Qiagen Catalog #51804-50** following the manufacturer's protocol (outlined below).

Note

Tip: If you are following the manufacturer's printed protocol, whenever the text says "vortex briefly" vortex for at least 5-10 seconds.

- 9 Check solution CD3. If solution CD3 has precipitated, heat solution to 60°C in a water bath until precipitate has dissolved.
- 10 To a microcentrifuge tube, add 1 mL of feces-PBS slurry from Sample Preparation step 7.



When transferring sample from the feces-PBS slurry pipette from the meniscus at the red arrow.



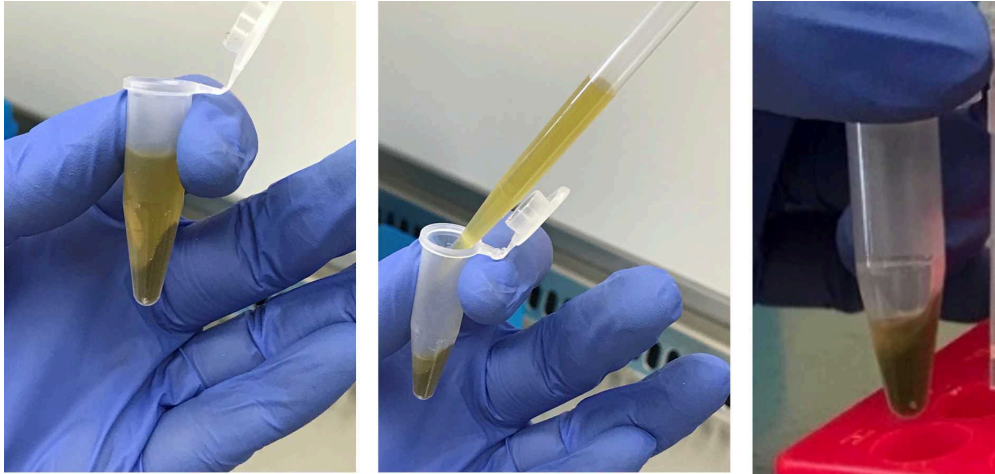
Pipetting the Feces-PBS slurry to a microcentrifuge tube.

- 11 Centrifuge at $15000 \times g$, Room temperature, 00:01:00

1m



- 12 Remove and carefully discard $800\ \mu\text{L}$ of the supernatant without disturbing the pellet.



Removing 800 uL of the supernatant.

- 13 Add $800\ \mu\text{L}$ Solution CD1 (Qiagen), and add $4\ \mu\text{L}$ VetMAX Xeno Internal Positive Control, and resuspend the fecal pellet.



VetMAX Xeno Internal Positive Control ND **Thermo Fisher**
Scientific Catalog #A29762/A29764

- 14 Transfer the fecal pellet that is resuspended in CD1 to a bead tube.



Bead tube from the manufacturer.

15 Secure the bead tubes horizontally on a

Equipment	
Vortex Adapter	NAME
Tube holder	TYPE
Qiagen (MoBio)	BRAND
13000-V1-24	SKU



Vortexer with Qiagen (MoBio) vortex adapter.

or secure the tubes on a flat-bed vortex pad with tape.

- 16 Vortex at **maximum speed** for  00:10:00 10 min.

10m

Note

Tip: If vortexing more than 12 tubes, add an additional 5 min.

- 17 Centrifuge at  15000 x g, Room temperature, 00:01:00

1m



- 18 Transfer  500-600 μL of the supernatant to a clean 2mL Microcentrifuge Tube.

Note

Note: Expect 500-600 μL of supernatant. The supernatant may still contain some stool particles.

- 19 Add  200 μL of CD2 and vortex for  00:00:05 .

5s

- 20 Centrifuge at  15000 x g, Room temperature, 00:01:00

1m

- 21 Avoiding the pellet, transfer up to  700 μL of supernatant to a clean 2 ml Microcentrifuge Tube

Note

Note: Expect 500-600 μL of supernatant.



22 Add  600 μL of Solution CD3 and vortex for  00:00:05

5s

23 Load  650 μL of lysate onto a MB Spin Column.



Spin Column

24 Centrifuge at  15000 x g, Room temperature, 00:01:00

1m



25 Discard the flow-through, place the MB Spin Column back into the same 2ml Collection Tube and repeat steps 24 and 25 to ensure that all of the supernatant has been loaded



onto the MB Spin Column.

26 Carefully place the MB Spin Column into a clean 2ml Collection Tube. Avoid splashing any flow-through onto the MB Spin Column.

27 Add  500 μL of Solution EA to the MB Spin Column.

28 Centrifuge at  15000 x g, 00:01:00

1m

29 Discard the flow through and place the MB Spin Column back into the same 2 ml Collection Tube.

30 Add  500 μL of Solution C5.

31 Centrifuge at  15000 x g, 00:01:00

1m

32 Discard the flow through and place the MB Spin Column in a clean 2 ml Collection Tube.

33 Centrifuge at  16000 x g, Room temperature, 00:02:00

2m

34 Carefully place the MB Spin Column into a new 1.5 ml Elution Tube

35 Add  60 μL of Solution C6 to the center of filter membrane

36 Centrifuge at  15000 x g, Room temperature, 00:01:00

1m

37 Discard the MB Spin Column. The DNA is now ready for PCR.

38

Note

If PCR will be performed within 1-5 days store the DNA at 4 °C . If PCR will be performed after 5 days post DNA isolation, store the DNA at -20 °C until ready to run PCR

PCR Method - Setting up and running the PCR reactions

- 39 The probe is labeled with FAM with Black Hole Quencher 1 (BHQ1) so detection parameters appropriate for FAM or SYBR should be used. The developing laboratory uses Applied Biosystems™ VetMAX™ Xeno™ Internal Positive Control DNA (ThermoFisher Scientific A29762 or A29764) and the VetMAX™ Xeno™ Internal Positive Control - VIC™ Assay (ThermoFisher Scientific A29765 or A29767) to monitor for PCR inhibition.

Note

Threshold is set at 0.2.

- 40 Create run on ABI 7500 Fast machine. Setting up the ABI 7500 Fast Thermocycler: Refer to the manual provided for running the machine. Reaction times are as follows in Table 1.

Equipment

7500 Fast Real-Time PCR System, desktop	NAME
Applied Biosystems	BRAND
4351107	SKU

	A	B	C	D	E	F
	Stage	Function	Temperature	Duration (sec)	Repeats	Optics

	A	B	C	D	E	F
		Reverse Transcription	50°C	300		
	Stage 1	Initial PCR Activation	95.0°C	20		Off
	Stage 2	Annealing	95.0°C	3	Repeats 45 times	Off
	Stage 2	Extension	60°C	60		On

Table 1a. PCR amplification conditions in ABI 7500 under standard conditions are as follows:

	A	B	C	D	E	F
	Stage	Function	Temperature	Duration (sec)	Repeats	Optics
		Reverse Transcription	50°C	300		
	Stage 1	Initial PCR Activation	95.0°C	20		Off
	Stage 2	Annealing	95.0°C	3	Repeats 45 times	Off
	Stage 2	Extension	60°C	60		On

Table 1b. PCR amplification conditions in ABI 7500 under FAST conditions are as follows:

Note

Passive reference dye should be set as ROX

- 41 Calculate the number of PCR reactions to be performed. Using Table 2 as a guide, determine the quantity of each reagent needed. Include a positive (*C. jejuni* ATCC 33291, ATCC 33292, or ATCC 33560) and negative (*E. coli* ATCC 25922) control and a "No Template" negative control.

For calculations, reference **Table 2.** PCR Worksheet for Reagent Volumes for Master Mix Preparation for *Campylobacter jejuni* PCR.



Table2 PCR Worksheet Campy Ma...

42 Combine reagents except for the DNA template in a 1.5ml or 2ml microcentrifuge tube and mix well.

43 Aliquot  18 μ L into each PCR tube or reaction well if using a plate.

44 Add samples and positive (*C. jejuni* ATCC 33291, ATCC 33292, or ATCC 33560 DNA) and negative (*E. coli* ATCC 25922 DNA) controls along with PCR inhibition controls



VetMAX Xeno Internal Positive Control ND Thermo Fisher
Scientific Catalog #A29762/A29764

;Lot_____).

45 Place Samples in thermocycler and perform run.

46 Following run, analyze the results. Results should be reported with a Ct value and interpretation.

47 Samples with a Ct value of 40 or less should be interpreted as positive. Samples with a Ct value >40 or an undetermined Ct value should be considered negative.

	A	B	C
	Target	CT Value	Interpretation
	gyrA	<=40	Detected
		>40	Not detected

Table 3. Interpretation of Ct Value



A	B
Oligo Name	5'-3' Sequence
gyrA-F	AAGATACGGTCGATTTTGTTC
gyrA-R	CTACAGCTATACCACTTGAACCATTTAATA
gyrA-P	5'-[FAM]TGATGGTTCAGAAAGCGAACCTGATGTTTT[BHQ1] 3'

Table 4. Primers and Probe